Performance Optimization of FAS-Assisted U2G Communications Using Reinforcement Learning

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***Abstract*—Fluid antenna systems (FAS) are emerging as a promising technology for sixth-generation (6G) networks due to their flexible and reconfigurable radiation characteristics. In this paper, we investigate the performance optimization of FAS-assisted unmanned aerial vehicle (UAV)-to-ground (U2G) communications. Unlike conventional static antenna configurations, the proposed framework leverages the unique ability of FAS to dynamically activate and adjust antenna ports, enabling optimization in U2G channels. A reinforcement learning approach based on an enhanced proximal policy optimization (PPO) algorithm, termed Memetic PPO, is introduced to efficiently explore the discrete FAS configuration space by integrating a local search mechanism. Key performance metrics, including channel capacity and modeling accuracy, are derived and analyzed. Simulation results demonstrate that the number of active ports and port spacing significantly affect both metrics, and the proposed Memetic PPO algorithm achieves superior configurations by effectively balancing these factors. These findings provide valuable insights for the design and optimization of FAS in practical UAV communication systems.**

***Index Terms*—Fluid antenna system, unmanned aerial vehicle, channel capacity, reinforcement learning, proximal policy optimization.**

I. Introduction

Despite the remarkable achievements of fifth-generation (5G), it still faces some shortcomings, such as an over-reliance on base stations and high costs. In contrast, sixth-generation (6G), as the next-generation communication technology, aims to integrate satellites to build a globally covered network and features low costs, holding promising development prospects. The fluid antenna system (FAS) is regarded as one of the key technologies for 6G and has demonstrated vigorous development momentum. The core concept of FAS lies in the reconfigurability of antenna shape and position, with similar ideas traceable back to the directionally reconfigurable waveguide antenna proposed in 1960. It was not until recent years that the concept of FAS was proposed in [1] and [2]. Essentially, FAS refers to any controllable fluid, conductive, or dielectric structure that can dynamically alter its shape and position to reconfigure key parameters such as gain, polarization, radiation pattern, and operating frequency, thereby enhancing signal transmission efficiency, spectrum utilization, and coverage. Currently, there exists substantial academic research on FAS. [3] indicated that under identical conditions, FAS exhibits higher secrecy capacity, lower outage probability, and superior energy efficiency compared to conventional antennas. Furthermore, several effective hardware implementations have emerged. For instance, the PB-FAS proposed in [4] offers low power consumption, low cost, and easy integration, making it suitable for large-scale smart city deployment. [5] found that in communication scenarios where base stations are equipped with fluid antennas, channel capacity can be enhanced by optimizing the antenna positions at both the base station and the user ends. [6] pointed out that when users are equipped with FAS, the scalability of the network can be improved by alleviating the base station's channel state information burden. [7] comprehensively presented the design, channel modeling, and experiments of an ER-FAS, the results demonstrated that their proposed ER-FAS performs effectively in both the near-field and far-field, thereby enhancing the overall spectral efficiency of the wireless communication system. Recent years have witnessed the rapid development of artificial intelligence (AI), and the deep integration of communication and AI aligns with the vision of 6G. [8] first introduced Bayesian regression into FAS and proposed S-BAR, a general, high-precision, low-pilot-overhead channel estimation method for FAS with strong robustness and adaptability.

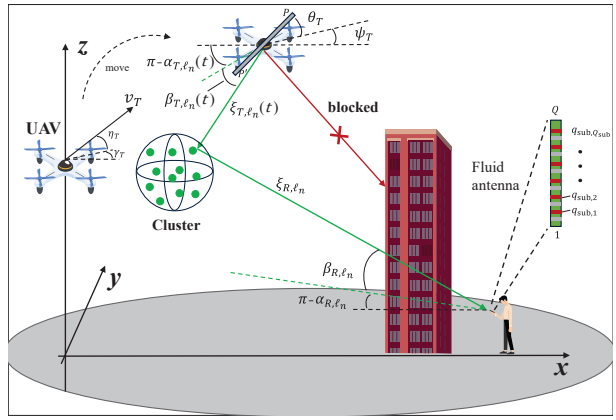

Fig. 1: Channel model of UAV-to-ground communication scenario assisted by FAS.

Although research on FAS is extensive, applying reinforcement learning (RL) to dynamic unmanned aerial vehicle (UAV)-to-ground (U2G) scenarios remains limited [8], [9]. To address the challenges of time-varying channels and the low exploration efficiency of baseline proximal policy optimization (PPO) in discrete action spaces, we propose an enhanced variant termed *S-MPPO*. This algorithm integrates a local

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search mechanism to refine action selection before execution, significantly accelerating convergence and uncovering superior configurations.

Notation: The lowercase, boldface lowercase, and boldface uppercase letters such as x, $\mathbf{x}$, and $\mathbf{X}$ stand for the scalars, vectors, and matries, respectively. $\|\cdot\|$ represents the Frobenius norm, $(\cdot)^{\mathrm{H}}$ denotes the conjugate transpose operation, and $\mathbb{E}[\cdot]$ denotes the expectation operator.

II. System Model

Consider a three-dimensional (3D) channel model for A2G communications as shown in Fig. 1, the UAV moves at a speed of v_T, and the azimuth and elevation angles of the direction of the motion are γ_T and η_T, respectively. In our model, the projection of midpoint of the UAV on the ground is defined as the coordinate origin, and the direction pointing to the receiver on the ground is the positive direction of the x-axis; the z-axis goes straight up and y-axis follows the right-hand rule. It is worth mentioning that the coordinate system does not move along with the UAV. Define H_0 and D_0 as the initial altitude of the UAV and the distance from the origin to the receiver, respectively.

On the UAV end, the azimuth and elevation orientation angles of the P omni-directional uniform linear array (ULA) antennas equipped on the UAV are respectively described as ψ_T and θ_T. The position of the p-th element in ULA can be expressed as

$$\mathbf{d}_p(t) = \mathbf{d}_T(t) + \mathbf{d}_{T,p}, \tag{1}$$

where $\mathbf{d}_T(t)$ denotes the time-varying position of the UAV, and $\mathbf{d}_{T,p}$ denotes the distance vector from the ULA midpoint to the p-th antenna. To be specific, they can respectively be expressed as

$$\mathbf{d}_T(t) = v_T t \begin{bmatrix} \cos\eta_T \cos\gamma_T \\ \cos\eta_T \sin\gamma_T \\ \sin\eta_T \end{bmatrix} + \mathbf{d}_{t_0}, \tag{2}$$

$$\mathbf{d}_{T,p} = k_p \delta_T \begin{bmatrix} \cos\theta_T \cos\psi_T \\ \cos\theta_T \sin\psi_T \\ \sin\theta_T \end{bmatrix}, \tag{3}$$

where $k_p = (P - 2p + 1)/2$, $\mathbf{d}_{t_0} = [0, 0, H_0]^{\mathrm{T}}$ and δ_T is the distance between adjacent antennas.

The receiver on the ground is equipped with a fluid antenna with a length of $W\lambda$ and Q ports, where λ is the carrier wavelength and $Q \gg Q_{\text{sub}}$, with Q_{sub} being the number of activated ports. The adjacent ports spacing of the fluid antenna Δ can be expressed as

$$\Delta = \frac{W\lambda}{Q-1}. \tag{4}$$

To characterize the dynamic port activation, we define a mapping relationship between the activated port index and the physical port index. Specifically, the q_{sub}-th activated port ($q_{\text{sub}} = 1, 2, \ldots, Q_{\text{sub}}$) corresponds to the q-th physical port ($q = 1, 2, \ldots, Q$) in the FAS through a mapping function $q = \mathcal{M}(q_{\text{sub}})$. Therefore, the position of the q_{sub}-th activated port relative to the global coordinate origin can be expressed as

$$\mathbf{d}_q = \mathbf{d}_R + k_q \Delta \begin{bmatrix} \cos\theta_R \cos\psi_R \\ \cos\theta_R \sin\psi_R \\ \sin\theta_R \end{bmatrix}, \tag{5}$$

where $k_q = (Q - 2\mathcal{M}(q_{\text{sub}}) + 1)/2$, ψ_R, θ_R are respectively the azimuth and elevation orientation angles of the FAS, and $\mathbf{d}_R = [D_0, 0, 0]^{\mathrm{T}}$ accounts for the position of receiver relative to the global coordinate origin.

The signal transmitted from the UAV propagates through non-line of sight (NLoS) components, undergoing scattering by clusters before arriving at the receiver. Due to UAV mobility, the angles of the scattering paths vary over time. Under the approximation, the angles of departure (AoD) for each path are modeled as fixed random variables. Specifically, for the n-th path in the ℓ-th cluster, the azimuth angle $\alpha_{T,\ell n}$ and elevation angle $\beta_{T,\ell n}$ are randomly generated from von Mises distributions:$\alpha_{T,\ell n} \sim \mathrm{VM}(\mu_{\alpha,\ell}, \kappa)$ and $\beta_{T,\ell n} \sim \mathrm{VM}(\mu_{\beta,\ell}, \kappa)$, where $\mu_{\alpha,\ell}$ and $\mu_{\beta,\ell}$ are the mean azimuth and elevation angles of the ℓ-th cluster, respectively, and κ is the concentration parameter.

We can represent the channel matrix for the proposed channel model as $\mathbf{H}^{\text{cluster}}(t,\tau)$, where τ denotes the path delay. Each element $h_{p,q_{\text{sub}}}(t,\tau)$ in $\mathbf{H}^{\text{cluster}}(t,\tau)$ stands for the complex channel impulse response (CIR) of the propagation link from the p-th element in ULA to the q_{sub}-th element in FAS, which can be given as

$$h_{p,q_{\text{sub}}}(t,\tau) = h^{\text{cluster}}_{p,q_{\text{sub}}}(t)\delta\left(\tau - \tau^{\text{cluster}}(t)\right), \tag{6}$$

where $\tau^{\text{cluster}}(t)$ is the path delay for the NLoS propagation link, and $h^{\text{cluster}}_{p,q_{\text{sub}}}(t)$ can be expressed as

$$\begin{aligned} h^{\text{cluster}}_{p,q_{\text{sub}}}(t) = \sum_{\ell\in L}\sum_{n=1}^{\ell_N} & e^{j\varphi_{\ell n}} \cdot e^{-j\frac{2\pi}{\lambda}(\xi_{T,\ell n}+\xi_{R,\ell n})} \\ & \times e^{j\frac{2\pi}{\lambda}k_p\delta_T\cos(\alpha_{T,\ell n}-\psi_T)\cos\beta_{T,\ell n}\cos\theta_T} \\ & \times e^{j\frac{2\pi}{\lambda}k_p\delta_T\sin\beta_{T,\ell n}\sin\theta_T} \\ & \times e^{j\frac{2\pi}{\lambda}k_q\Delta\cos(\alpha_{R,\ell n}-\psi_R)\cos\beta_{R,\ell n}\cos\theta_R} \\ & \times e^{j\frac{2\pi}{\lambda}k_q\Delta\sin\beta_{R,\ell n}\sin\theta_R} \\ & \times e^{j\frac{2\pi}{\lambda}v_T t\cos(\alpha_{T,\ell n}-\gamma_T)\cos\beta_{T,\ell n}\cos\eta_T} \\ & \times e^{j\frac{2\pi}{\lambda}v_T t\sin\beta_{T,\ell n}\sin\eta_T}, \end{aligned} \tag{7}$$

where $\{\varphi_{\ell n}\}_{n=1,2,\ldots,N}$ represents the random phase, assumed to be independent and uniformly distributed as $\varphi_{\ell n} \sim \mathrm{U}[-\pi,\pi)$. The terms $\xi_{R,\ell n} = \|\mathbf{d}_{\ell_n} - \mathbf{d}_R\|$ denote distances between the receiver and the n-th path in the ℓ-th cluster. Furthermore, α_{R,ℓ_n} and β_{R,ℓ_n} are angles of arrival.

For U2G wireless communication systems, channel capacity is a crucial indicator of the achievable spectral efficiency, which is given by

$$C = \log_2\left(\det\left(\mathbf{I}_{Q_{\text{sub}}} + \frac{\rho_{\text{SNR}}}{P}\overline{\mathbf{H}^{\text{cluster}}}(t,\tau)\overline{\mathbf{H}^{\text{cluster}}}^{\mathrm{H}}(t,\tau)\right)\right), \tag{8}$$

where $\mathbf{I}_{Q_{\text{sub}}}$ denotes the identity matrix of size $Q_{\text{sub}} \times Q_{\text{sub}}$, ρ_{SNR} is the signal-to-noise ratio (SNR), and $\overline{\mathbf{H}^{\text{cluster}}}(t,\tau)$ represents the normalized channel matrix.

III. Spatial-based Memetic PPO Model

To address the FAS configuration challenge in dynamic U2G communications, we propose the spatial-based memetic PPO (S-MPPO) framework. The core of this framework is the PPO algorithm, a policy-gradient method that ensures stable training through clipped surrogate objectives. Unlike standard reinforcement learning approaches that rely on blind stochastic exploration, S-MPPO embeds physical channel priors directly into the decision-making loop, enabling the agent to exploit the spatial correlation of the FAS ports through a localized memetic search. This integrated architecture maintains the theoretical stability of PPO while significantly enhancing sample efficiency for dynamic FAS configuration. S-MPPO formulates the problem as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, R, \gamma)$, with the specific details elaborated in the following parts.

The state space $\mathcal{S}$ is designed to capture the dynamic U2G channel characteristics and FAS configuration. Given the complexity of the channel model described in Section II, we select key physical quantities that directly influence system performance:

$$\mathbf{s} = [\mathbf{d}_T(t), \Delta H, \nabla C, \Delta\phi, \mathbf{Q}, C] \in \mathcal{S}, \tag{9}$$

where $\mathbf{d}_T(t)$ represents the instantaneous position vector of the UAV at time t as defined in (2), which captures the spatial dynamics of the communication link. $\Delta H = \|\mathbf{H}^{\text{cluster}}(t) - \mathbf{H}^{\text{cluster}}(t-1)\|_F$ denotes the channel variation rate caused by UAV mobility. ∇C denotes the capacity gradient, defined as the mean difference in channel capacity between adjacent ports, and represents the potential capacity improvement achievable through port reconfiguration. $\Delta\phi$ represents the phase variation rate, defined as the mean absolute phase difference between adjacent ports, and characterizes the spatial variation of multipath interference across the fluid antenna. $\mathbf{Q} \in \{0,1\}^N$ is a binary port activation mask where $Q_q = 1$ indicates that the q-th physical port is active while $Q_q = 0$ denotes it is inactive. C is the channel capacity computed as (8), serving as the primary performance metric for the FAS configuration. It is worth emphasizing that a fundamental characteristic of fluid antenna systems is the strong spatial correlation existing among densely distributed ports. Owing to the compact and quasi-continuous structure of FAS, neighboring ports experience highly similar propagation paths, which leads to correlated channel responses. We explicitly incorporate this spatial correlation of FAS into the state representation through ΔH, ∇C, and $\mathbf{m}$.

The action space $\mathcal{A}$ is a discrete space defined by the following catalog:

$$\mathcal{A} = \{a_{\text{none}}\} \cup \{a_{\text{replace}}(q, \delta) \mid q \in \{0, 1, \ldots, Q-1\}\ \}, \tag{10}$$

where a_{none} denotes the no-operation action that maintains the current port configuration, and $a_{\text{replace}}(q, \delta)$ represents a replacement action that deactivates port q and activates port $q + \delta$. In order to direct the agent's focus towards optimizing the spatial positions of FAS ports, we deliberately exclude the option of increasing the number of active ports from the action space $\mathcal{A}$. Experimental findings indicate that if such an option were available, the agent would develop a strong tendency to simply activate more ports rather than optimizing their placement. This behavior would contradict the fundamental design philosophy of FAS, which emphasizes achieving superior performance through intelligent spatial reconfiguration of a limited set of ports, rather than merely relying on port quantity. The reward function R is designed to guide the agent towards optimizing the spatial configuration of FAS ports, which can be defined as:

$$R(\mathbf{s}_t, a_t) = k_1 \cdot G_{a_t} + k_2 \cdot C_t + k_3 \cdot A_t^{\text{ULA}}, \tag{11}$$

where $G_{a_t} = C_{t+1} - C_t^{(\text{none})}$ is the capacity improvement achieved by executing action a_t compared to the hypothetical scenario of taking the no-operation action. This term directly encourages the agent to exploit spatial correlation and seek better local port placements. C_t is the channel capacity achieved by the current active port set at time step t. Including this absolute value provides a stable performance baseline and prevents excessive exploration. $A_t^{\text{ULA}} = C_t - C_{\text{ULA}}$ is the performance advantage of the current FAS configuration over a baseline ULA with an equivalent number of RF chains, which incentivizes the agent to surpass conventional antenna array performance as much as possible. k_1, k_2, and k_3 are non-negative weighting coefficients that balance the emphasis between immediate gain, absolute performance, and baseline advantage, respectively. This composite reward structure ensures that the agent is not only rewarded for making beneficial adjustments but also for maintaining high overall capacity and outperforming traditional architectures.

Algorithm 1 S-MPPO for Dynamic FAS Configuration

Initialize policy π_θ and value function V_ϕ with random parameters
for each episode **do**
  Initialize UAV trajectory and sample multi-path clusters for $\mathbf{H}^{\text{cluster}}(t)$
  Set initial FAS port activation mask $\mathbf{Q}_0$
  for $t = 0, \ldots, T-1$ **do**
    Observe current state $\mathbf{s}_t$
    Sample candidate policy action $a_t^{\text{PPO}} \sim \pi_\theta(\mathbf{s}_t)$
    Spatially-aware Refinement:
    1. Construct neighborhood $\mathcal{N}(a_t^{\text{PPO}})$ based on action displacement δ
    2. Identify $q_{\text{sub}}^{\min} \in \mathbf{Q}_t$ with minimum $\|h_{q_{\text{sub}}}\|_2$
    3. Identify strongest idle ports to determine S-M swap action $a_t^{\text{S-M}}$
    4. Form augmented candidate set $\mathcal{A}_t^{\text{cand}}$
    Select refined action a_t^*
    Execute a_t^*, update mask to $\mathbf{Q}_{t+1}$, and receive reward $R(\mathbf{s}_t, a_t^*)$
    Store transition $(\mathbf{s}_t, a_t^*, R(\mathbf{s}_t, a_t^*), \mathbf{s}_{t+1})$ in $\mathcal{D}$
  end for
  Compute $\hat{A}_t$ and returns for all t in $\mathcal{D}$
  Update θ and ϕ by maximizing $\mathcal{L}^{\text{CLIP}}(\theta)$ and minimizing value loss
  Clear buffer $\mathcal{D}$
end for

At each time step t, the agent observes the state $\mathbf{s}_t \in \mathcal{S}$ that includes the characteristics of the UAV and the spatial correlation of FAS. The agent then samples a candidate action $a_t^{\text{PPO}} \sim \pi_\theta(\mathbf{s}_t)$, providing global exploration over the discrete configuration space. However, directly executing this action may be inefficient due to the combinatorial nature of the problem. This challenge can be addressed by leveraging a key property of FAS: owing to the compact and quasi-continuous structure of fluid antenna systems, the channel responses across adjacent ports are strongly spatially correlated. As a result, the capacity landscape over the port domain exhibits local smoothness, where neighboring configurations tend to yield similar performance. This property motivates the incorporation of a spatially-aware refinement mechanism.

Specifically, instead of directly executing a_t^{PPO}, we construct an augmented candidate action set $\mathcal{A}_t^{\text{cand}}$ via a synergistic process of global policy mapping and local heuristic refinement:

$$\mathcal{A}_t^{\text{cand}} = \{a_t^{\text{PPO}}\} \cup \mathcal{N}(a_t^{\text{PPO}}) \cup \{a_t^{\text{S-M}}\}, \tag{12}$$

where $\mathcal{N}(a_t^{\text{PPO}})$ denotes the neighborhood of the policy's suggestion, exploiting the local smoothness of the capacity landscape. The term $a_t^{\text{S-M}}$ represents a spatial-based memetic (S-M) swap action, to be specific, it identifies the weakest active port $q_{\text{sub}}^{\min} = \arg\min_{q_{\text{sub}} \in Q_{\text{sub}}} \|h_{q_{\text{sub}}}\|_2$ based on the instantaneous channel power spectrum and attempts to replace it with the strongest available idle port, effectively leveraging the spatial correlation of FAS. Among all candidate actions in the augmented set, the final executed action a_t^* is determined by a localized greedy maximizer:

$$a_t^* = \arg \max_{a' \in \mathcal{A}_t^{\text{cand}}} C(\mathbf{H}(t), \text{apply}(a', \mathbf{Q}_t)), \tag{13}$$

where $C(\cdot)$ denotes the spectral efficiency achieved under the configuration resulting from applying action a' to the current mask $\mathbf{Q}_t$. By executing a_t^*, the agent transitions to the next state $\mathbf{s}_{t+1}$ and receives the composite reward $R(\mathbf{s}_t, a_t^*)$ which emphasizes the spatial configuration gain.

The objective of the S-MPPO agent is to learn an optimal policy π_θ that maximizes the expected discounted cumulative reward over the UAV's trajectory

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} \gamma^t R(\mathbf{s}_t, a_t^*)\right], \tag{14}$$

where $\gamma \in (0, 1]$ is the discount factor. To ensure monotonic and stable policy improvement while incorporating the results of the memetic refinement, we employ the clipped surrogate objective of PPO:

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\hat{A}_t, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right], \tag{15}$$

where $\rho_t(\theta) = \frac{\pi_\theta(a_t^*|\mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(a_t^*|\mathbf{s}_t)}$ represents the probability ratio between the current and old policies for the refined action a_t^*. The advantage $\hat{A}_t$ is computed using Generalized Advantage Estimation (GAE) to balance bias and variance in the dynamic U2G environment.

By storing the refined transitions $\{(\mathbf{s}_t, a_t^*, R_t, \mathbf{s}_{t+1})\}$ in the experience replay buffer $\mathcal{D}$, the policy π_θ effectively learns to approximate the behavior of an idealized, channel-aware spatial optimizer. Crucially, the memetic refinement operates as a decision-support layer during interaction, allowing the agent to benefit from high-quality experience trajectories while preserving the convergence guarantees and theoretical stability of the PPO algorithm. This synergistic integration allows S-MPPO to efficiently exploit the inherent spatial smoothness of the fluid antenna array, leading to superior spectral efficiency in non-stationary communication scenarios.

IV. Results and Analysis

The experiment employs parameters for U2G communications: carrier frequency $f_c = 28$ GHz, UAV initial altitude $H_0 = 20$ m, initial horizontal distance $D_0 = 50$ m, number of UAV antennas $P = 4$, total FAS ports $Q = 50$, UAV antenna spacing $\delta_T = \lambda/2$, UAV velocity $v_T = 10$ m/s, UAV motion azimuth angle $\gamma_T = 45°$, UAV motion elevation angle $\eta_T = 15°$, FAS aperture length $W = 3\lambda$, the scattering environment consists of $L = 2$ clusters with 10 paths per cluster, following a von Mises distribution with an environment factor $\kappa = 2.0$. The signal-to-noise ratio (SNR) is set to $\rho_{\text{SNR}} = 2$ dB. Training utilizes the PPO algorithm with the following hyperparameters: learning rate 3×10^{-4}, steps per update 1024, batch size 256, epochs per update 10, discount factor $\gamma = 0.99$, generalized advantage estimation parameter $\lambda = 0.95$, clipping parameter $\epsilon = 0.1$, entropy coefficient 0.05, and value function coefficient 0.5. The policy network comprises two hidden layers with 128 neurons each, using ReLU activation functions. Training proceeds for 200,000 timesteps, with the environment's episode length set to $T = 30$ steps to capture the UAV mobility dynamics. Model checkpoints are saved based on evaluation performance.

Fig. 2 illustrates the average performance gain as a function of the number of active ports Q_{sub} for the proposed S-MPPO, the baseline PPO, and a fixed ULA configuration. From the simulation results, it can be observed that both FAS-based optimization methods significantly outperform the conventional fixed ULA. This advantage stems from the spatial adaptability of the fluid antenna, which allows the agent to navigate the quasi-continuous port space and circumvent deep fading in the dynamic U2G channel. Moreover, the proposed S-MPPO consistently maintains a superior performance margin over the baseline PPO, and notably, as Q_{sub} increases, the performance gap between S-MPPO and the baseline PPO widens. This occurs because a larger Q_{sub} increases the dimensionality of the discrete configuration space, making it increasingly difficult for standard stochastic exploration to locate the global optimum. In contrast, S-MPPO leverages a spatially-aware memetic refinement mechanism to prune suboptimal configurations and exploit local channel smoothness, thereby converging to higher-quality solutions. It is worth mentioning that the diminishing return in performance gain as Q_{sub} grows suggests that while increasing the number of RF chains improves performance, the intelligent spatial placement of a few ports is more critical than merely increasing the port count, validating the core design philosophy of the proposed S-MPPO framework.

Fig. 3 compares the performance gain of the proposed S-MPPO and the baseline PPO over the fixed ULA across a

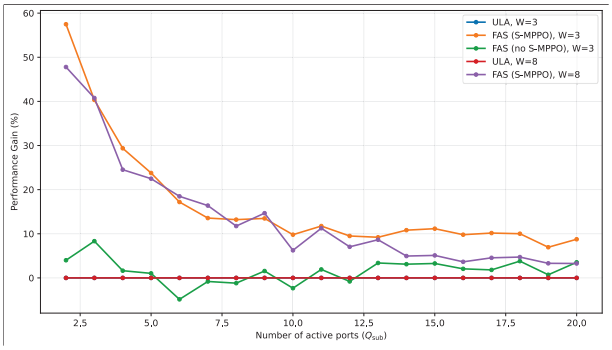

Fig. 2: Performance gain vs Q_{sub}.

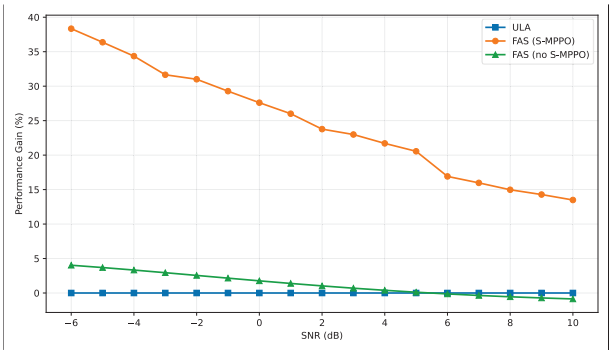

Fig. 3: Performance gain vs SNR, where W=3 and $Q_{\text{sub}} = 5$.

wide range of SNR, specifically from -6 dB to 10 dB. As expected, the gains of both schemes decrease monotonically with the SNR, since the absolute channel quality of the U2G link approaches saturation in the high-SNR regime, leaving less room for relative improvement. However, the proposed S-MPPO consistently achieves higher gains than the baseline PPO throughout the entire SNR regime. Even in low-SNR conditions, S-MPPO maintains a noticeable lead. This demonstrates that the spatially-aware memetic refinement effectively identifies high-gain ports even when the channel state is obscured by significant noise, thereby maximizing the limited link budget. Furthermore, it is observed that as the SNR increases, the gain gap between S-MPPO and the baseline PPO narrows in absolute terms, yet the relative advantage remains robust. The persistent superiority over the baseline PPO confirms that the integration of the spatially-aware memetic refinement mechanism allows the agent to reach configurations closer to the theoretical capacity limit, proving the efficacy of combining global policy learning with local physical-layer heuristics in non-stationary U2G environments.

To evaluate the system performance under a joint optimization framework, the weighting factors in (**??**) are empirically assigned as $r_c = 0.2$ and $r_{\text{error}} = -0.1$. These coefficients are normalized to enhance numerical stability during the training process, with the relative magnitudes reflecting a prioritized focus on maximizing channel capacity while simultaneously penalizing modeling inaccuracies. Under this multi-objective setting, the baseline PPO converges to a configuration of $W = 15.0$ and $Q_{\text{sub}} = 30$, whereas the memetic PPO identifies a more refined solution of $W = 14.0$ and $Q_{\text{sub}} = 30$. The

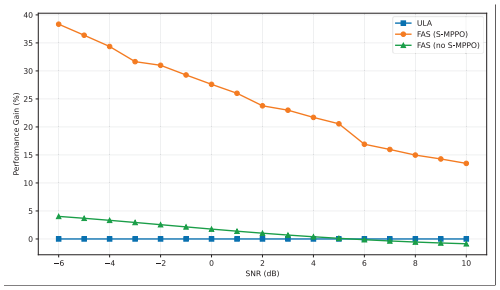

Fig. 4: Comprehensive score with different FAS configurations

functional relationship between the comprehensive score and the FAS parameters is illustrated in Fig. 4. The simulation results demonstrate that the composite utility increases monotonically with the number of activated ports. Although an increase in Q_{sub} inevitably exacerbates the modeling error [10], the chosen weighting strategy ensures that the gains in spectral efficiency dominate the overall performance metric. Specifically, the positive reward for capacity outweighs the penalty incurred by modeling deviations, thereby favoring configurations with high Q_{sub}. Furthermore, the configuration attained by the memetic PPO ($W = 14.0$) yields a superior comprehensive score compared to the baseline PPO's result ($W = 15.0$). This performance margin is attributable to the hybrid optimization architecture of memetic PPO, which synergistically combines stochastic gradient-based policy updates with deterministic local search to navigate the complex multi-objective landscape more effectively. The discrepancy in the identified optimal antenna length suggests that memetic PPO is capable of discovering a more balanced equilibriumłwhere a moderate antenna length maintains high capacity while effectively managing system complexity. This underscores the algorithm's robustness in capturing nuanced trade-offs that remain elusive to the standard PPO framework.

V. Conclusion

This paper has investigated the performance optimization problem for a FAS-assisted U2G communication system using reinforcement learning. The proposed framework can adaptively configure FAS parameters, such as antenna length and the number of active ports, in dynamic UAV scenarios, exploring the potential of data-driven methods for enhancing U2G communication performance. Through comprehensive simulation and comparative analysis, the effectiveness of the PPO algorithm and its memetic variant has been validated. It has been demonstrated that the number of active ports and the fluid antenna length significantly impact the trade-off between channel capacity and modeling error. Specifically, while increasing the number of active ports leads to higher channel capacity, it simultaneously elevates the modeling error, emphasizing the requirement for an intelligent configuration strategy. Moreover, the memetic PPO variant is shown to achieve superior comprehensive scores and better convergence stability compared to the standard PPO by integrating deterministic local search. Finally, it should be pointed out that applying this RL-based framework to multi-user interference

scenarios and incorporating practical constraints like energy efficiency are key research directions in our future research.

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